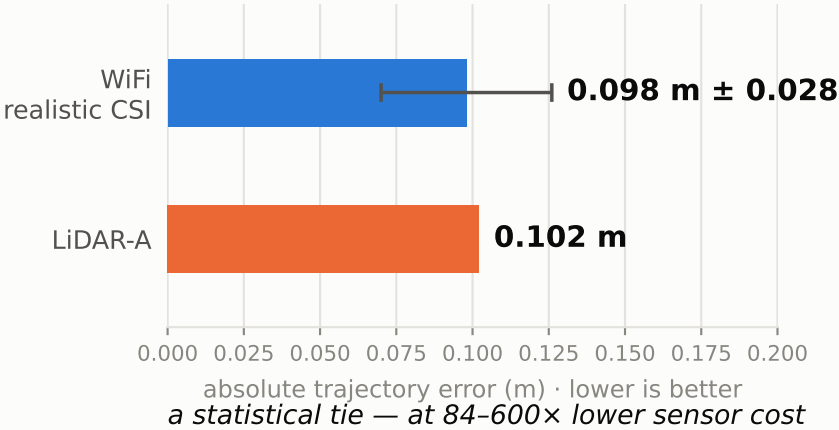


Can ambient WiFi replace radar or LiDAR as a SLAM front-end?

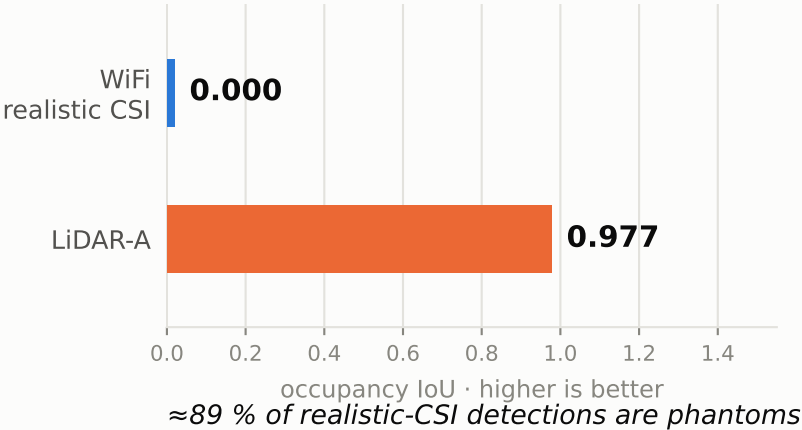
Ray-traced sub-7 GHz WiFi on a moving vehicle, scored against real-LiDAR-anchored baselines and reproduced on \$30 of commodity silicon. Controlled scene.

WiFi · 5.2 GHz LiDAR radar · 77 GHz

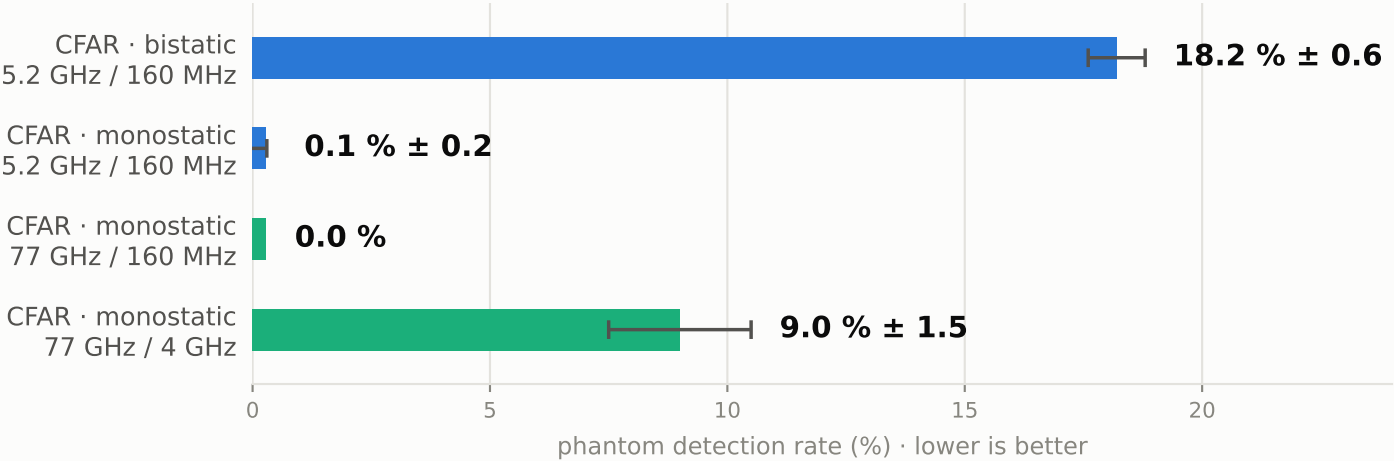
✓ LOCALIZATION works



✗ MAPPING hits a ceiling



Why? The ceiling is GEOMETRY, not the carrier



- ↓ **geometry**
phantoms all but vanish
- ↓ **carrier**
no change
- ↓ **10× bandwidth**
phantoms return

ON \$30 OF SILICON

two ESP32-S3 boards capture real HT40 CSI — 128 subcarriers, 114 active